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# =====
# PYTHON LIBRARY FILES IMPLEMENTATION
# HOSPITAL DATASET - 15 RECORDS
# =====

import pandas as pd
import matplotlib.pyplot as plt

# -----
# CREATE HOSPITAL DATASET
# -----

data = {
    "Patient_ID": [
        101, 102, 103, 104, 105,
        106, 107, 108, 109, 110,
        111, 112, 113, 114, 105
    ],
    "Name": [
        "Arun", "Priya", "Karthik", "Divya", "Rahul",
        "Sneha", "Vijay", "Anu", "Surya", "Meena",
        "Ajay", "Kavya", "Ravi", "Nisha", "Rahul"
    ],
    "Age": [
        25, 45, 32, 60, 50,
        28, 67, 39, 55, 42,
        70, 34, 48, 63, 50
    ],
    "Gender": [
        "Male", "Female", "Male", "Female", "Male",
        "Female", "Male", "Female", "Male", "Female",
        "Male", "Female", "Male", "Female", "Male"
    ],
    "Department": [
        "Cardiology", "Neurology", "Orthopedic", "Cardiology", "General",
        "Neurology", "Cardiology", "Orthopedic", "General", "Neurology",
        "Cardiology", "General", "Orthopedic", "Cardiology", "General"
    ],
    "Disease": [
        "Heart Disease", "Migraine", "Fracture", "Heart Disease", "Fever",
        "Migraine", "Heart Disease", "Fracture", "Fever", "Migraine",
        "Heart Disease", "Fever", "Fracture", "Heart Disease", "Fever"
    ],
    "Stay_Days": [
        5, 3, 7, 8, 2,
        4, 10, 6, 3, 4,
        9, 2, 5, 7, 2
    ],
    "Bill_Amount": [
        45000, 18000, 32000, 65000, 12000,
        22000, 80000, 28000, 15000, 20000,
        72000, 13000, 30000, 55000, 12000
    ]
}

df = pd.DataFrame(data)

print("===== ORIGINAL HOSPITAL DATASET =====")
print(df)

print("\nDataset Shape:", df.shape)

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# =====
# 1. FILTERING DATA
# =====

print("\n\n===== 1. FILTERING DATA =====")

# Patients with bill greater than 30000

filtered_data = df[df["Bill_Amount"] > 30000]

print("\nPatients with Bill Amount > 30000:")
print(filtered_data)

# =====
# 2. FILTERING DATA WITH MULTIPLE CONDITIONS
# =====

print("\n\n===== 2. MULTIPLE CONDITIONS =====")

# Patients who:
# Age > 50 AND Stay_Days > 5

multiple_filter = df[
    (df["Age"] > 50) &
    (df["Stay_Days"] > 5)
]

print("\nPatients with Age > 50 AND Stay Days > 5:")
print(multiple_filter)

# Another example:
# Cardiology patients with bill > 40000

cardiology_filter = df[
    (df["Department"] == "Cardiology") &
    (df["Bill_Amount"] > 40000)
]

print("\nCardiology patients with Bill > 40000:")
print(cardiology_filter)

# =====
# 3. SORTING DATA BY BOTH ORDERS
# =====

print("\n\n===== 3. SORTING DATA =====")

# Ascending order

ascending = df.sort_values(
    by="Bill_Amount",
    ascending=True
)

print("\nBill Amount - Ascending Order:")
print(ascending[["Patient_ID", "Name", "Bill_Amount"]])

# Descending order

descending = df.sort_values(
    by="Bill_Amount",
    ascending=False
)

print("\nBill Amount - Descending Order:")
print(descending[["Patient_ID", "Name", "Bill_Amount"]])

# =====
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# 4. REMOVING DUPLICATE DATA
# =====

print("\n\n===== 4. REMOVING DUPLICATE DATA =====")

print("\nNumber of rows BEFORE removing duplicates:")
print(len(df))

# Find duplicate rows

duplicates = df[df.duplicated()]

print("\nDuplicate rows:")
print(duplicates)

# Remove duplicates

df_no_duplicates = df.drop_duplicates()

print("\nNumber of rows AFTER removing duplicates:")
print(len(df_no_duplicates))

# =====
# 5. AGGREGATION
# =====

print("\n\n===== 5. AGGREGATION =====")

# Average hospital bill

average_bill = df["Bill_Amount"].mean()

print("\nAverage Bill:", average_bill)

# Maximum bill

maximum_bill = df["Bill_Amount"].max()

print("Maximum Bill:", maximum_bill)

# Minimum bill

minimum_bill = df["Bill_Amount"].min()

print("Minimum Bill:", minimum_bill)

# Average bill by department

department_bill = df.groupby(
    "Department"
)["Bill_Amount"].mean()

print("\nAverage Bill by Department:")
print(department_bill)

# Total bill by department

department_total = df.groupby(
    "Department"
)["Bill_Amount"].sum()

print("\nTotal Bill by Department:")
print(department_total)

# Number of patients in each department

patient count = df.groupby(
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    "Department"
)["Patient_ID"].count()

print("\nNumber of Patients by Department:")
print(patient_count)

# Complete aggregation

summary = df.groupby("Department").agg(
    Average_Bill=("Bill_Amount", "mean"),
    Maximum_Bill=("Bill_Amount", "max"),
    Minimum_Bill=("Bill_Amount", "min"),
    Average_Stay=("Stay_Days", "mean")
)

print("\nDepartment Summary:")
print(summary)

# =====
# 6. DATA VISUALIZATION
# =====

# -----
# 6.1 LINE PLOT
# -----

print("\n\n===== 6.1 LINE PLOT =====")

plt.figure(figsize=(8, 5))

plt.plot(
    df["Patient_ID"],
    df["Bill_Amount"],
    marker="o"
)

plt.title("Patient ID vs Hospital Bill")
plt.xlabel("Patient ID")
plt.ylabel("Bill Amount")

plt.grid(True)

plt.show()

# -----
# 6.2 BAR CHART
# -----

print("\n\n===== 6.2 BAR CHART =====")

plt.figure(figsize=(8, 5))

department_total = df.groupby(
    "Department"
)["Bill_Amount"].sum()

plt.bar(
    department_total.index,
    department_total.values
)

plt.title("Total Hospital Bill by Department")
plt.xlabel("Department")
plt.ylabel("Total Bill Amount")

plt.xticks(rotation=30)

plt.show()

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# -----
# 6.3 HISTOGRAM
# -----

print("\n\n===== 6.3 HISTOGRAM =====")

plt.figure(figsize=(7, 5))

plt.hist(
    df["Age"],
    bins=5,
    edgecolor="black"
)

plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Number of Patients")

plt.show()

# -----
# 6.4 SCATTER PLOT
# -----

print("\n\n===== 6.4 SCATTER PLOT =====")

plt.figure(figsize=(7, 5))

plt.scatter(
    df["Stay_Days"],
    df["Bill_Amount"]
)

plt.title("Hospital Stay vs Bill Amount")
plt.xlabel("Stay Days")
plt.ylabel("Bill Amount")

plt.grid(True)

plt.show()

# -----
# 6.5 PIE CHART
# -----

print("\n\n===== 6.5 PIE CHART =====")

department_count = df["Department"].value_counts()

plt.figure(figsize=(7, 7))

plt.pie(
    department_count.values,
    labels=department_count.index,
    autopct="%1.1f%"
)

plt.title("Patients by Department")

plt.show()

# =====
# FINAL DATASET
# =====

print("\n\n===== FINAL DATASET =====")

print(df_no_duplicates)
```



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===== ORIGINAL HOSPITAL DATASET =====
Patient_ID  Name  Age  Gender  Department  Disease  Stay_Days  \
0           101  Arun  25   Male  Cardiology  Heart Disease    5
1           102  Priya 45  Female  Neurology  Migraine        3
2           103  Karthik 32   Male  Orthopedic  Fracture        7
3           104  Divya 60  Female  Cardiology  Heart Disease    8
4           105  Rahul 50   Male   General    Fever          2
5           106  Sneha 28  Female  Neurology  Migraine        4
6           107  Vijay 67   Male  Cardiology  Heart Disease   10
7           108  Anu   39  Female  Orthopedic  Fracture        6
8           109  Surya 55   Male   General    Fever          3
9           110  Meena 42  Female  Neurology  Migraine        4
10          111  Ajay 70   Male  Cardiology  Heart Disease    9
11          112  Kavya 34  Female   General    Fever          2
12          113  Ravi  48   Male  Orthopedic  Fracture        5
13          114  Nisha 63  Female  Cardiology  Heart Disease    7
14          105  Rahul 50   Male   General    Fever          2

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Bill_Amount
0           45000
1           18000
2           32000
3           65000
4           12000
5           22000
6           80000
7           28000
8           15000
9           20000
10          72000
11          13000
12          30000
13          55000
14          12000

```

Dataset Shape: (15, 8)

===== 1. FILTERING DATA =====

Patients with Bill Amount > 30000:

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Patient_ID  Name  Age  Gender  Department  Disease  Stay_Days  \
0           101  Arun  25   Male  Cardiology  Heart Disease    5
2           103  Karthik 32   Male  Orthopedic  Fracture        7
3           104  Divya 60  Female  Cardiology  Heart Disease    8
6           107  Vijay 67   Male  Cardiology  Heart Disease   10
10          111  Ajay 70   Male  Cardiology  Heart Disease    9
13          114  Nisha 63  Female  Cardiology  Heart Disease    7

Bill_Amount
0           45000
2           32000
3           65000
6           80000
10          72000
13          55000

```

===== 2. MULTIPLE CONDITIONS =====

Patients with Age > 50 AND Stay Days > 5:

```

Patient_ID  Name  Age  Gender  Department  Disease  Stay_Days  \
3           104  Divya 60  Female  Cardiology  Heart Disease    8
6           107  Vijay 67   Male  Cardiology  Heart Disease   10
10          111  Ajay 70   Male  Cardiology  Heart Disease    9
13          114  Nisha 63  Female  Cardiology  Heart Disease    7

Bill_Amount
3           65000
6           80000
10          72000
13          55000

```

Cardiology patients with Bill > 40000:

```

Patient_ID  Name  Age  Gender  Department  Disease  Stay_Days  \
0           101  Arun  25   Male  Cardiology  Heart Disease    5
3           104  Divya 60  Female  Cardiology  Heart Disease    8
6           107  Vijay 67   Male  Cardiology  Heart Disease   10
10          111  Ajay 70   Male  Cardiology  Heart Disease    9
13          114  Nisha 63  Female  Cardiology  Heart Disease    7

```

```

13      114  Nisha    55  Female  Cardiology  Heart Disease  /

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```

      Bill_Amount
0      45000
3      65000
6      80000
10     72000
13     55000

```

===== 3. SORTING DATA =====

Bill Amount - Ascending Order:

	Patient_ID	Name	Bill_Amount
4	105	Rahul	12000
14	105	Rahul	12000
11	112	Kavya	13000
8	109	Surya	15000
1	102	Priya	18000
9	110	Meena	20000
5	106	Sneha	22000
7	108	Anu	28000
12	113	Ravi	30000
2	103	Karthik	32000
0	101	Arun	45000
13	114	Nisha	55000
3	104	Divya	65000
10	111	Ajay	72000
6	107	Vijay	80000

Bill Amount - Descending Order:

	Patient_ID	Name	Bill_Amount
6	107	Vijay	80000
10	111	Ajay	72000
3	104	Divya	65000
13	114	Nisha	55000
0	101	Arun	45000
2	103	Karthik	32000
12	113	Ravi	30000
7	108	Anu	28000
5	106	Sneha	22000
9	110	Meena	20000
1	102	Priya	18000
8	109	Surya	15000
11	112	Kavya	13000
4	105	Rahul	12000
14	105	Rahul	12000

===== 4. REMOVING DUPLICATE DATA =====

Number of rows BEFORE removing duplicates:

15

Duplicate rows:

	Patient_ID	Name	Age	Gender	Department	Disease	Stay_Days	Bill_Amount
14	105	Rahul	50	Male	General	Fever	2	12000

Number of rows AFTER removing duplicates:

14

===== 5. AGGREGATION =====

Average Bill: 34600.0

Maximum Bill: 80000

Minimum Bill: 12000

Average Bill by Department:

Department

Cardiology 63400.0

General 13000.0

Neurology 20000.0

Orthopedic 30000.0

Name: Bill_Amount, dtype: float64

Total Bill by Department:

Department

Cardiology 317000

General 52000

Neurology 60000


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neurology      60000
Orthopedic     90000
Name: Bill_Amount, dtype: int64

```

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Number of Patients by Department:
Department
Cardiology      5
General         4
Neurology       3
Orthopedic      3
Name: Patient_ID, dtype: int64

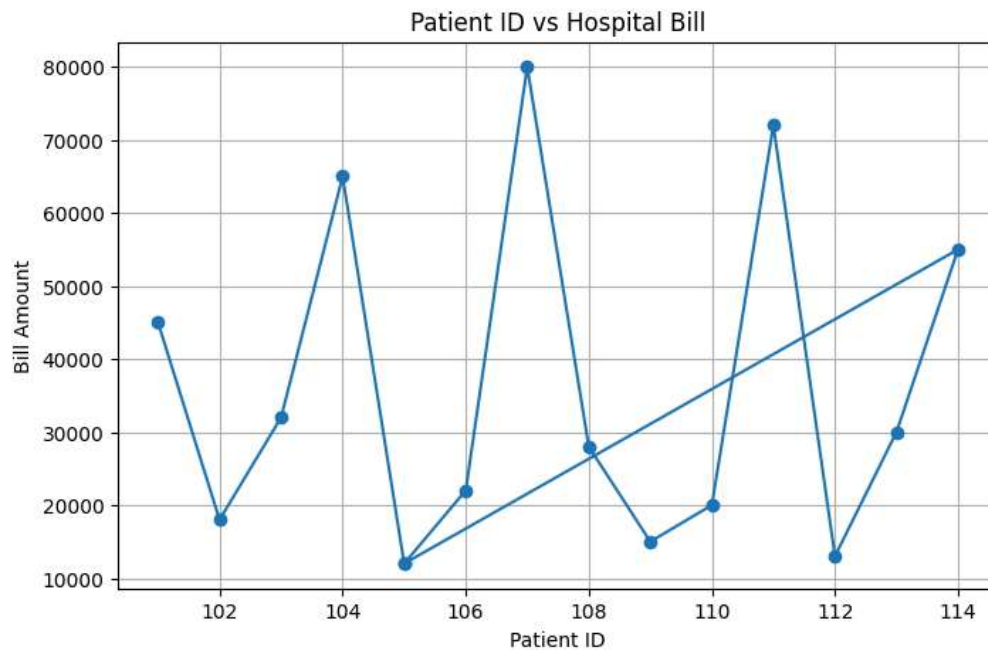
```

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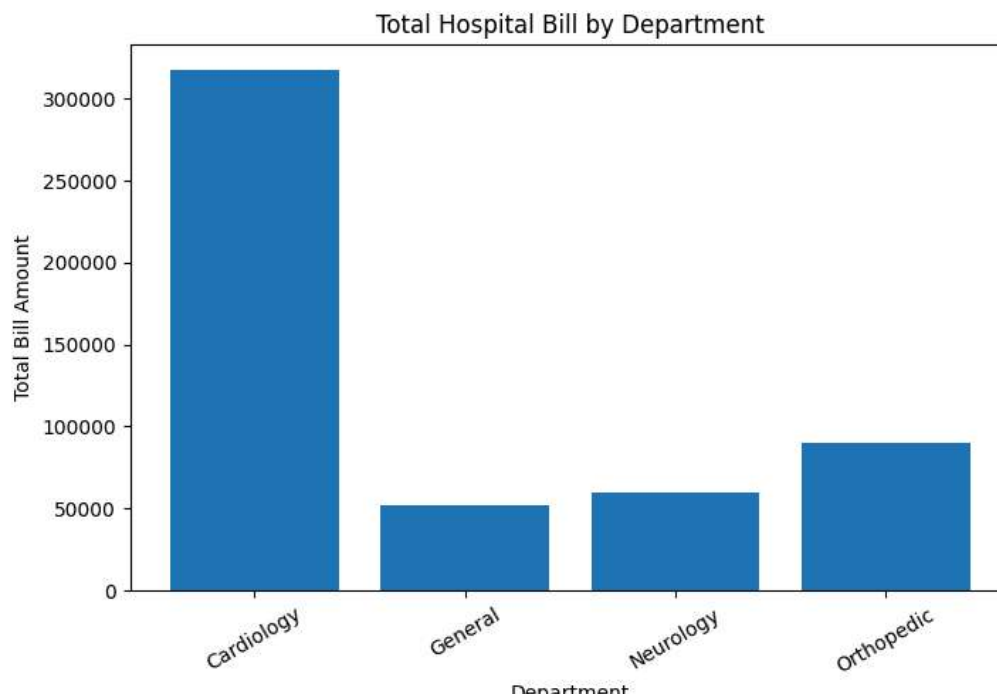
Department Summary:
Average_Bill  Maximum_Bill  Minimum_Bill  Average_Stay
Department
Cardiology    63400.0         80000         45000         7.800000
General       13000.0         15000         12000         2.250000
Neurology     20000.0         22000         18000         3.666667
Orthopedic    30000.0         32000         28000         6.000000

```

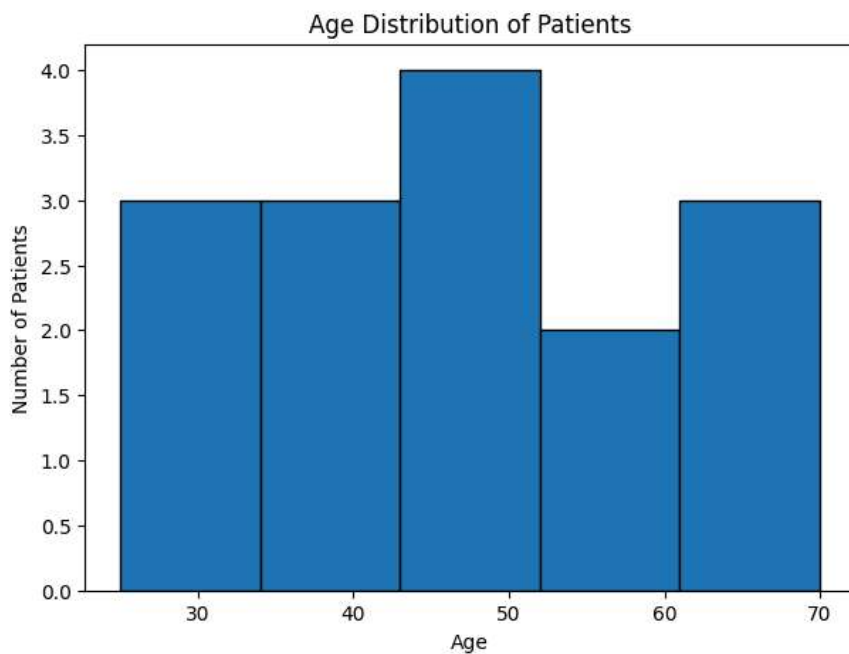
===== 6.1 LINE PLOT =====



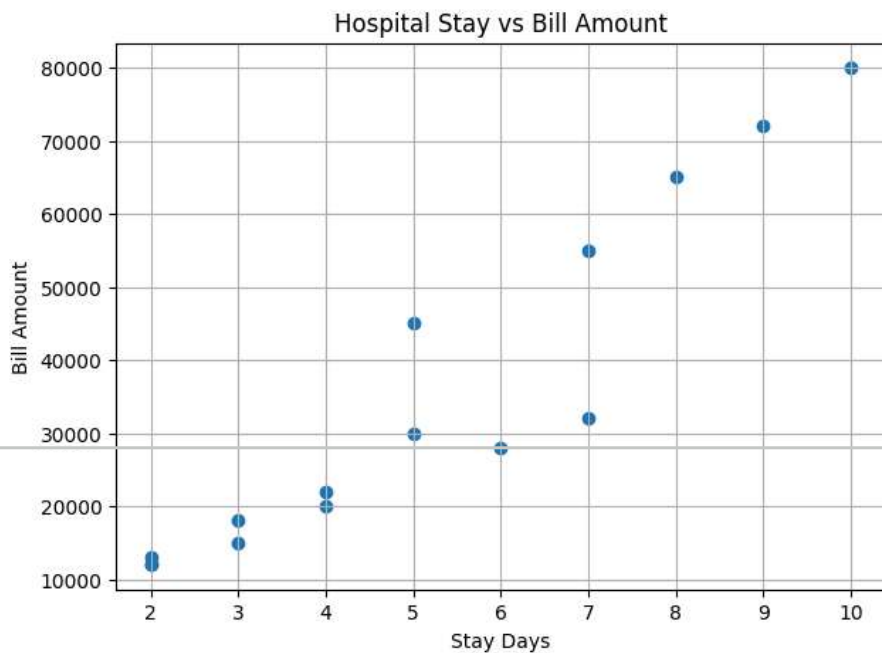
===== 6.2 BAR CHART =====



===== 6.3 HISTOGRAM =====



===== 6.4 SCATTER PLOT =====



===== 6.5 PIE CHART =====

